

Did I find the cheatcode of DevOps?

Or am I just yapping about AI?



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- Nerd
- Developer
- Loves automation
- Gamer
- AuDHD

AI AGAIN??!

Yes - or maybe?



04:12 AM







"Hey Manny! this is very early, I hope you got some sleep
I've been hard at work since I woke you up 4.3 minutes ago..."



■ ERROR

ALERT: checkout-api error rate > 20% for 5 minutes

Severity: Critical

■ Grafana

```
HTTP 500 rate: 2% -> 34%  
HTTP 503 rate: 0% -> 18%  
Checkout success rate: 97.8% -> 41.2%
```




"So I looked into the logs for checkout-api and found these"

```
04:12:04 ERROR [http-nio-8080-exec-42] c.company.checkout.CheckoutController
Failed to complete checkout for cartId=cart_98217 userId=user_44123

java.sql.SQLTransientConnectionException: HikariPool-1 - Connection is not available, request timed out after 30000ms.
    at com.zaxxer.hikari.pool.HikariPool.createTimeoutException(HikariPool.java:696)
    at com.zaxxer.hikari.pool.HikariPool.getConnection(HikariPool.java:197)
```

"And also these"

```
04:12:06 WARN [checkout-api] Circuit breaker 'orders-db' recorded a slow call
04:12:07 WARN [checkout-api] Circuit breaker 'orders-db' changed state from CLOSED to HALF_OPEN
04:12:09 ERROR [checkout-api] Order creation failed: timeout acquiring DB connection
04:12:11 WARN [checkout-api] Retrying createOrder attempt=3 cartId=cart_98217
04:12:14 ERROR [checkout-api] Checkout failed after retries
```

"So I started looking into if the **Database connection pool could be exhausted**"



"But when I checked the database it looked normal"

```
CPU: 38%  
Connections: 61% of max  
IOPS: normal  
Replication lag: 0.2s
```

"but at the same time application log started showing"

```
WARN [checkout-api] OrderRepository.createOrder took 5021ms  
WARN [checkout-api] retrying order creation after transient failure
```



"So I started to see if there was any changes done to the system recently"

```
Change: Update CoreDNS forwarding rules for internal zones
Owner: Platform team
Time: 03:47 AM
Reason: Prepare migration from legacy internal DNS to Route 53 Resolver
```

"there was a change in the CoreDNS ConfigMap:"

```
prod.internal:53 {
forward . 10.12.0.10
}
```

```
internal:53 {
forward . 10.12.0.10
}
```



"Then, there was a autoscaler event performing a routine capacity rebalacing at 04:02 AM"

"It seems the existing pods kept their connection, but the new ones tried reaching `orders-db.prod.internal`"

"... and that does no longer exist"



"CoreDns logs confirms my analysis: "

```
04:12:03 [ERROR] plugin/errors: 2 orders-db.prod.internal. A: read udp 10.42.3.18:53422->10.12.0.10:53: i/o timeout
04:12:04 [ERROR] plugin/errors: 2 orders-db.prod.internal. AAAA: read udp 10.42.3.18:52191->10.12.0.10:53: i/o timeout
04:12:05 [INFO] 10.42.7.91:44931 - 54822 "A IN orders-db.prod.internal. udp 51 false 512" SERVFAIL
04:12:05 [INFO] 10.42.7.91:42019 - 32415 "AAAA IN orders-db.prod.internal. udp 51 false 512" SERVFAIL
04:12:07 [INFO] 10.42.8.14:39012 - 9814 "A IN orders-db.prod.internal. udp 51 false 512" NXDOMAIN
```



"To check I tried this: `kubectrl exec -it checkout-api-7f6b8f9b9c-jrx2m -- nslookup orders-db.prod.internal`"

"...and got different responses at different times"

```
** server can't find orders-db.prod.internal:  
SERVFAIL
```

```
** server can't find orders-db.prod.internal:  
NXDOMAIN
```

```
Name: orders-db.prod.internal  
Address: 10.88.14.27
```



"I suggest to roll back the CoreDNS ConfigMap manually, and notify the Platform Team"

```
prod.internal:53 {  
  errors  
  cache 30  
  forward . 10.12.0.10  
}
```

```
kubectl rollout restart deployment/coredns -n kube-system
```

"Do you want me to apply?"

Yes/No [Y/n]:

Y



"Config map updated. Let me check..."

```
kubectl exec -n prod deploy/checkout-api -- dig orders-db.prod.internal A  
> orders-db.prod.internal. 60 IN A 10.88.14.27
```

"Let me also make sure we redeploy all the order-apis"

```
kubectl rollout restart deployment/checkout-api -n prod
```




"Everything seems to be back to normal"

"I am sorry I had to wake you up, but at least it's still just 4:23 AM, so I hope you can get a few more hours of sleep tonight!"

"I'll send a short notification to the Platform Engineer on call, and start preparing the Incident Report. I'll have that ready for you when you get up"

"Good night!"

zzz



"Good morning, I have prepared an incident report for you!"

Incident Report

■ Outage Dec. 25, 2025 - 04:12 AM

Start	End	Severity
04:12 AM	04:23 AM	Critical

▣ Impact

~40% of checkout requests failed or timed out. Users could browse and add items to cart, but many could not complete purchases

▣ Symptoms, Investigation and timeline

Sid magically found out

Time	Event
03:47	CoreDNS config change applied to generalize internal forwarding rules
03:48	CoreDNS reloads successfully

...

▣ Root case

A recent CoreDNS configuration change broke forwarding for the `prod.internal` DNS zone. New checkout pods could not reliably resolve the database hostname, causing connection pool exhaustion.

▣ Follow-up actions

Notify Platform Team

■ Executive Summary

It looked like the database. It was DNS wearing a fake mustache

It was DNS

It is always DNS

Even when it's not DNS, DNS is nearby - wearing sunglasses

Incident Copilot

■ Triage and Remediation Assistant

- Correlate signals across observability (logs/metrics/traces), recent deploys, config changes, feature flags.
- Suggest “likely causes” with confidence and show evidence.
- Recommend next diagnostic steps (queries to run, dashboards to open).
- Writes the initial report

■ Human impact

- Reduce cognitive load
- Reduce time to fix the problem
- Writes the boring report for me

■ Guardrail

"No autonomous remediation" - explicit approval for any changes

Ops Concierge

■ A Single Front Door

- "Who owns service X?"
- "What changed in prod in the last hour?"
- "Is this alert actionable or known-noisy?"
- "Open an incident at SEV2 and invite the right people"

▒ Human impact

Reduces interruptions, lowers tribal knowledge dependence, speeds up coordination

▒ Best pattern

Make sure all answers have links to real human-made documentation + suggested actions

▒ Our day-to-day hero!



"I have a new minor release in production. Can you perform a risk assessment?"

Application	Magic Wand Navigator
Current prod tag	1.0.69
New Tag	1.1.0



"Of course, my dear Sir"

Change Risk Summarizer

■ Risk aware release-assistant

- For a proposed deploy, summarize: what changed, touched components, blast radius, dependencies.
- Classify change risk (routine vs elevated) based on heuristics (critical path, config-only, DB migration, large diff, new dependency).
- Check readiness: runbook updated? dashboards exist? rollback plan? on-call aware? SLO impact?
- Draft the change record automatically (for regulated environments).

■ Human impact

Makes change management faster and less bureaucratic

■ Guardrail

Risk score, change record etc is **advisory**. A human puts their name on it



1:15 PM

Incident Ticket

Ticket details

Severity:	SEV-3
Impact:	Site operational, high reputation/business risk
Description:	"Discrete Packaging" metadata breach

Expected Packaging Slip

"ACME - Home massager"

Actual Packaging Slip

"Dual-Node Auto-Scaler"

"Chaos Monkey Multi-Speed"





"Hey Team!

I created a temporary Slack channel for this - 'War Room - INC1069'

Let's keep all the chat about this incident in there"



"So I started to see if there was any changes done to the system recently"

```
Change: Minor change in shipping-manifest-api
Owner:  Jack Junior
Time:   12:47 AM
Reason: Change to improve formatting of international addresses
```

 Diff

```
1 ...
2 label.Address = addresses.Format(customer.Address)
3 ...
4 label.Description = append(label.Description, orderline.DiscreteAlias)
5 ...
```

```
1 ...
2 label.Address = addresses.FormatInternational(customer.Address)
3 ...
4 label.Description = append(label.Description, orderline.ProductName)
5 ...
```



* Taking notes of chat *

* Watches what changes is being done *

Incident Report

■ Incident, December 28., 2025 - 01:15 PM

Start	End	Severity
01:15 PM	01:53 AM	SEV-3 - high

▣ Impact

Discrete packages stopped being discrete. Printed product names on shipping label.

▣ Symptoms, Investigation and timeline

Manual SEV-3 report from fulfilment center

Time	Event
12:47 AM	Shipping label update - for handling of international addresses
01:16 PM	Chat: Team starts discussing the issue. Alice notices the code change
01:27 PM	Git: Code fix submitted by Bill - rullback of one code line
01:29 PM	Prod: Rollback Successful
01:57 PM	Chat: Team confirms incident is fixed
...	

▣ Root case

Junior developer had misunderstood why we had discrete shipping labels, and changed the label because he thought is was an error

▣ Follow-up actions

Notify Developer team - tell them to go easy on the junior

Website Announcement

🚧 ⚠️ Notice: We experienced some unexpected downtime (and not the relaxing kind)

Hey everyone,

Earlier today, our systems experienced a **sudden surge**. While we usually pride ourselves on handling **heavy loads** and delivering guaranteed uptime, one of our backend systems got **a little too excited** and **released** a minor update **prematurely**.

To maintain our famously strict "discreet shipping" standards (because we know you prefer your personal business to stay undercover), we had to briefly hit the pause button on our warehouse operations to ensure **nothing slipped out into the open**.

Our engineering team has rolled back the changes, applied the necessary patches, and verified that everything is running smoothly and quietly once again. We are now working double-time to ensure your **much-anticipated stress relief** arrives exactly as intended: **in a plain, totally boring brown box**.

We sincerely apologize for any **friction** this caused to your delivery timeline. To make up for it, please use the promo code **GOODVIBES15** for 15% off your next order.

Stay relaxed, The Engineering Team

Incident Coordinator Asssistant

Automate incident timeline capture

- When was it reported
- What was the severity and why?
- What lead up to this
 - Prior changes?

Draft customer/internal updates in agreed templates.

"Unforseen circumistances led to..."

Track action items live

- Alice investigating DB
- Bob rolling back

Post-incident Review

■ PIR drafting

- Draft incident narrative from chat + alerts + deploy timeline +++
- Extract contributing factors
 - missing alerts
 - risky change
 - unclear ownership
- Suggest follow-up tasks
- Identify recurring themes across incidents
 - "40% of incidents came after redeploy of ..."

■ Why?

PIRs are often not made, or is low quality

■ Guardrail

Keep PIR blameless, no "faults" only "contributing factors"

Alert-noise reduction

Have our agents just "living" in our production environment

■ De-noising Assistant

- Cluster alerts to identify duplicates/noise sources
- Recomend alert rule changes
- Detect "alarm storms"

■ Human impact

Reduces alert fatigue, improve trust in monitoring

■ Guardrails

Require review before modifying alert rules

Standard remediation

This require some maturity!

■ The Fixer

- "Allowed remediation tasks" based on earlier experiences
- Human-in-the-loop to confirm the fix
- Create standard messages to Teams, Slack, etc... to inform users

■ Human impact

Use less time on simple fixes

■ Guardrail

Always have a Human-in-the-loop unless you are 100% sure it's a safe remediation

Practical Guardrails

■ Non-negotiable Guardrails for All Use Cases

▒ Citations and traceability

Answers should link to dashboards, tickets, policies, commits...

▒ Least privilege + RBAC

- Read-only by default
- Write actions gated

▒ Human-in-the-loop approvals

Especially for production changes

Practical Guardrails

■ Non-negotiable Guardrails for All Use Cases

▒ Audit logs

Every action and the prompt/context that triggered it

▒ Safe failure mode

If uncertain, ask clarifying questions or escalate. **Never guess!**

▒ Data handling rules

Don't leak secrets, PII or incident-sensitive details into prompts/logs

▒ Clear ownership

Someone owns the bot, just like any other service

QUESTIONS?

Bonus question

■ Why is the bot called Sid in my examples?

▒ What is my name?

Manfred, goes by Manny

▒ Who do I share name with?



Thank you



